TFPT — Frontier items

η_B, m_p/m_e, the Koide relation, dark matter and quantum gravity — the honest status, not forced onto the ladder

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What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of η_B, m_p/m_e, the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **six short documents**, best read in order:

1. `introduction` — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
6. `tfpt_horizon_readouts` — one seam constant as the universal horizon thermal code.

You are reading document #5.

Contents

Ground rule

These five are exactly the items previously flagged "*not on a clean action/ladder yet — do not force.*" This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [I]/[P]/[A] are applied throughout. (All numbers machine-verified.)

Hard rule on the frontier (F_{frontier})

Koide, η_B, the axion relic scale and m_p/m_e are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied. They are deliberately typed interfaces, not structural holes: each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on a named, not-yet-performed calculation (the source→pole QFT shift; the flavored-leptogenesis Boltzmann solve; the near-hilltop relic integral with anharmonic/string corrections; the QCD/EW cross-sector matching). They must never be quoted as primitive TFPT outputs.

Why this is principled, not a gap: these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [`verification/v35_quark_qcd_boundary.py`]

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

Route A — from the closed baryon fraction. The seed already fixes the baryon *fraction* (Paper 3 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \tfrac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \qquad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10}\,\Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \qquad \eta_B = 6.09 \times 10^{-10}} \qquad (\text{observed } 6.1 \times 10^{-10}).$$

So η_B is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

Route B — leptogenesis interface (Paper 6, [I]). The closed branch also fixes the leptogenesis inputs: the heavy scale $M_R \approx 1.3 \times 10^{15}$ GeV, the CP phases (δ_{CKM}, δ_{CP}^{ν}, both closed), the reheating T_R, and the seam-fixed initial phase. The numerical η_B then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent cross-check of Route A. **Honest caveat:** that Boltzmann network (the washout/efficiency κ_f, the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.

Honest status of η_B — strictly typed [P]

The two statements must be kept apart and not interchanged:

$$\boxed{\eta_B \text{ as a cosmological readout from the closed } \Omega_b h^2 \text{ is } \textit{determined } (6.1\times 10^{-10})}$$

$$\boxed{\eta_B \text{ as a fundamental } \textit{compiler power} \text{ is } \textit{not} \text{ closed}}$$

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So η_B is [P]— determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

2 Proton/electron ratio m_p/m_e — a cross-sector ratio

m_e is *closed* on the $\varphi_0^{\rm ret}$-ladder ($\hat{m}_e = \frac{v_{\rm geo}\pi}{\sqrt{2}}\frac{16}{7}(\varphi_0^{\rm ret})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times\Lambda_{\rm QCD}$, and $\Lambda_{\rm QCD}$ arises by dimensional transmutation from α_s running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single $\varphi_0^{\rm ret}$ power: $1/(\varphi_0^{\rm ret})^2 = 353.7$ and $1/(\varphi_0^{\rm ret})^3 = 6651$ bracket 1836 with no clean rung in between.

Honest status of m_p/m_e [A]

Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed m_e), exactly like the dimensionful EW/QCD masses m_W, m_Z. But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [A] — not forced.

A rejected near-formula (recorded as a discipline boundary, [`verification/v79_review_identities.py`]). The combination $|W(D_5)| - \dim\Lambda^3(SU(9)) + N_{\rm fam}\lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected.* Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim\Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and m_p is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; m_p/m_e stays [A]. (The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q}c_3^6 \approx 6.10\times 10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — η_B stays [P], the downstream readout above.)

3 The Koide relation — near 2/3, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\rm ret})^5, \frac{4}{3}(\varphi_0^{\rm ret})^3, \frac{7}{6}(\varphi_0^{\rm ret})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\rm TFPT} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\ldots, \qquad \frac{2}{3} = 0.66667, \qquad \text{deviation } -0.33\%.$$

The measured pole-mass value is famously $Q_{\rm PDG} = 0.66666$. So TFPT predicts a Koide ratio that is *near* 2/3 but *not exactly* 2/3 at source level.

Honest status of Koide [P]

$Q_{\rm TFPT} = 0.664$ is a *prediction*, 0.33% below 2/3. Exact 2/3 would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force 2/3.

New: the *target* 2/3 has a compiler reading $|\mathbb{Z}_2|/N_{\rm fam}$

The E_8-slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly

$$Q_\star = \frac{|\mathbb{Z}_2|}{N_{\rm fam}} = \frac{2}{3},$$

the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\rm fam} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus "democratic 2/3 from $(|\mathbb{Z}_2|, N_{\rm fam})$, realised to 0.33% by the explicit masses" — promoted from a bare near-miss to a reading with a target. [P]

Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ [P]

The source value sits just below 2/3, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\rm source} = 0.6644638, \qquad Q_\ell^{\rm source} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only 1.3×10^{-5}. So the *conjecture* $Q_\ell^{\rm pole} \approx Q_\ell^{\rm source} + \varphi_0/|W(A_3)|$ would land essentially on the democratic 2/3. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [`verification/v25_frontier_conjectures.py`]

Sharper conjecture: a multiplicative seed transfer $u \to \frac{53}{54}u$ [P]

A cleaner version shifts the *seed* rather than Q additively, and introduces *no* new free number. Keep the lepton formula $m_e{:}m_\mu{:}m_\tau = \frac{16}{7}u^5{:}\frac{4}{3}u^3{:}\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u_{\rm pole}^{(\ell)} = \Big(1 - \tfrac{1}{2N_{\rm fam}^3}\Big)u = \tfrac{53}{54}\,u, \qquad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2|\,N_{\rm fam}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. 2/3 to within -5.7×10^{-7} — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is* $u \mapsto u\,(1 - \frac{1}{2N_{\rm fam}^3})$ (a renormalised-seed effect from the leptonic self-energy or the A_3 family-Weyl averaging). Status remains [P] until that transfer is derived. [`verification/v37_plucker_anchor.py`]

The Koide target and the QFT transfer gap are the *same* quotient [I]/[P]

The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible C_6 transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\boxed{\lambda_2 = \left(\tfrac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \qquad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\mathrm{fam}}}, \quad |R^+(A_3)| = 6}.$$

The Koide democratic target, the QFT mass-gap eigenvalue λ_2 (hence $\Delta = 6\log\frac{3}{2} = -\log\lambda_2$), and the A_3 positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [P], gap eigenvalue [I], the connection an [I]-audit. This does not "solve" Koide, but it explains *why* the near-value $2/3$ recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

Sharpening: the source→pole map is *forced* by that gap [I]/[P] ([verification/v82_koide_attractor_splitting.py])

The previous box pairs the Koide target with $\lambda_2 = (2/3)^6$; [verification/v82_koide_attractor_splitting.py] turns the pairing into a *mechanism*. On the anchor double cover (Paper 2) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once μ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$ with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \to 2$ ($x \to -2/3$) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value $2/3$, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [P]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto $2/3$.)

Relaxation toy: basin, exact rate, and two honest negatives [I]/[N] ([verification/v93_koide_relaxation_toy.py])

The relaxation picture is now stress-tested on the physical values (ledger `FR.KOIDE.05`). **Basin lemma [I] (new):** under the canonical dictionary $q = N_{\mathrm{fam}}\,Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [I]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [I]/[N]:** the exact ladder gives $Q_{\mathrm{src}} = 0.6644638$, i.e. $\rho_{\mathrm{src}} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ($24 = |W(A_3)|$); PDG pole masses give $Q_{\mathrm{pole}} = 0.6666645$, the branch point hit to 3.3×10^{-6}. **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\mathrm{pole}}/\rho_{\mathrm{src}}) = 2.84$ — *not* an integer, so the literal discrete iteration pole $= F^n(\mathrm{source})$ is *excluded*; if the v82 map drives the transfer at all, the missing [P] object is a *continuous-time* generator $\exp(t\log F)$; (ii) the 0.79% mismatch of ρ_{src} vs $-\varphi_0/24$ is far above ladder precision, so "exactly $\varphi_0/24$" remains a conjecture, not an identity. The open question is no longer "why 2/3?" but "what is the continuous transfer generator, and why is its flow time ~ 2.8 periods?".

Flow time: canonical generator, data-fragility, and a conditional m_τ [I]/[N]/[P] ([verification/v99_koide_flow_time.py])

Three sharpenings of the box above (ledger `FR.KOIDE.06`); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [I].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\rm fam}}(q-2)(q-5) = \frac{\Delta}{N_{\rm fam}}\det B(q)}, \qquad \Delta = 6\log\tfrac{3}{2} \ \ \text{(the gap)},$$

whose time-1 map *is* the Möbius map F (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [P] object is therefore precisely a QFT mechanism producing *gap* $\times$ *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of t is data-fragile [N].** Because $\rho_{\rm pole} = -2.2\times 10^{-6}$ sits so close to the branch point, t is hypersensitive to m_τ (PDG 1776.93 ± 0.09 MeV): $t(-1\sigma) = 2.35$, $t(\text{central}) = 2.84$, and Q crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t \to \infty$) — within $+0.5\sigma$, t passes *every* integer ≥ 3. So the exclusion of pole $= F^n(\text{source})$ holds only at the central value. **(3) Conditional discrete steps [P] (registry untouched; not predictions of record):** n=2 is *excluded* (-2.9σ, [N]); n=3=$N_{\rm fam}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \qquad (+0.14\sigma;\ \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while n=4 (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau)\sim 0.01$ MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t = \ln(M_{\rm scal}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to 10^{-4} — and is *destroyed* by negative controls (the same hypersensitivity makes *any* W in a wide range land within 0.1σ); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([P], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

4 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (Paper 6, Thm. "determinant-line axion interface", [I]). Two of its inputs are already *closed*:

$$\text{initial misalignment} \quad \theta_i = \pi(1-\varphi_{\rm seam}(\alpha_\star)) = 2.974 \text{ rad} = 170.4^\circ,$$

and the domain-wall number $N_{\rm DW}$ winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\rm mis}(f_a, m_a, \theta_i) + \Omega_a^{\rm str}(f_a, N_{\rm DW}),$$

and Paper 6 states that the minimal closed branch has *dominant axion-like dark matter.*

New: the E_8 scan rules out a rival WIMP candidate

The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16,4)\oplus(\overline{16},\overline{4})$ is the *same* matter spinor as in the $D_5\times A_3$ construction — there is *no spare E_8 singlet* left over to play a WIMP. Every E_8 direction is accounted for by the carrier $\times$ family content. So dark matter cannot be a new E_8 representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [I]

Honest status of dark matter [P]/[A]

The candidate is *identified* (the axion) and its misalignment angle $\theta_i = 170°$ is *closed* [I]. The relic abundance still needs the decay constant f_a and mass m_a (interface data set by the M_R/Λ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate + misalignment fixed, scale f_a pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near M_R.)

Conjecture: a determinant-line axion scale $f_a = M_{\rm scal}/128$ [P]

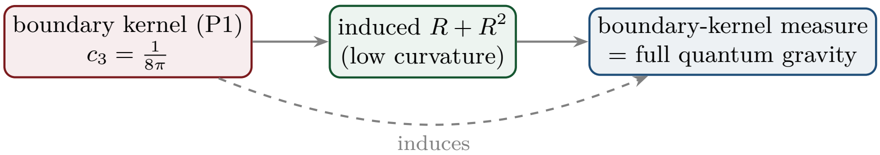
A natural compiler scale for the decay constant is the scalaron mass over the full carrier × deck multiplicity:

$$f_a = \frac{M_{\rm scal}}{2\,\dim S^+\,|\mu_4|} = \frac{M_{\rm scal}}{128}, \qquad M_{\rm scal} = c_3^{7/2} M_{\rm Pl} \approx 3.06 \times 10^{13}\,\text{GeV},$$

giving $f_a \approx 2.39 \times 10^{11}$ GeV and, via the standard QCD-axion relation $m_a \simeq 5.70\,\mu\text{eV}\,(10^{12}\,\text{GeV}/f_a)$, a mass $m_a \approx 23.8\,\mu$eV. This is consistent with the closed near-hilltop misalignment $\theta_i = 170.4°$, but it is a *cosmological scenario* (sensitive to anharmonic / string corrections near the hilltop), not a closed hit. [`verification/v25_frontier_conjectures.py`]

5 Full quantum gravity — induced from the boundary kernel

The normalisation $c_3 = 1/(8\pi)$ *is* the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The R^2/scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\rm Pl}$ *relative to* Λ, so G_N is read out *after* selecting the physical Λ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([I]/[P], [`verification/v60_lambda_metrology_branch.py`]) — a metrology readout, not a from-nothing prediction.



Honest status of quantum gravity — physical sector closed, ambient gap-decoupled [F]/[A]

"Full quantum gravity beyond the R^2 sector" is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\operatorname{Tr} f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + R^2 *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/M_{\rm Pl}^2 = 6(4\pi)^2/f_0$ with the TFPT closure c_3^7 fixing the cutoff moment f_0. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the un-built ambient, so under those assumptions the admissible-sector measure is closed. **What remains [A]:** the ambient metric measure — the projective limit (G6) — is the $G_{\rm metric}$ research gate; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [verification/v36_spectral_action_g2.py]

Update (QBL chain, v108–v113): the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank $= c$: 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[L]** (see `tfpt_1` and `tfpt_research_contracts`). [verification/v113_quasifree_kernel.py]

Summary

Item	Status	Honest handle / what is missing
η_B	**[P]**	downstream readout $= 6.1\times10^{-10}$ from closed $\Omega_b h^2$; leptogenesis is an interface (Boltzmann)
m_p/m_e	**[A]**	cross-sector (QCD scale / closed m_e); scheme-level, not a compiler power
Koide Q	**[P]**	computed $Q = 0.664$, 0.33% from 2/3; near, not exact — not tuned
dark matter	**[P]**/**[A]**	determinant-line axion candidate fixed, $\theta_i = 170°$ closed; relic scale f_a, m_a open
quantum gravity	**[F]**/**[A]**	$R + R^2$ heat-kernel grounded (G2); physical sector gap-decoupled; only ambient limit (G6) open

Marker key: ***[I]*** *exact identity ·* ***[L]*** *Lie/lattice theorem ·* ***[F]*** *formalised ·* ***[N]*** *numerical fixed point ·* ***[P]*** *physical/conditional ·* ***[A]*** *axiom / open.*

Net (honest). Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1 \times 10^{-10}$ via the closed Ω_b; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs 2/3). One is structural-only and left open (m_p/m_e as a cross-sector ratio); quantum gravity has been sharpened — its $R + R^2$ form is now heat-kernel grounded (G2) and its physical sector is gap-decoupled (G5), leaving only the ambient projective limit (G6) as a mathematical-completeness item. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

Appendix: computational verification

These are the *frontier* items: most carry **[A]** (open) or **[P]** (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is $\Omega_b=(1-\frac{1}{4\pi})\varphi_0^{\text{ret}}$ (whence the η_B order) in `verification/v7_gravity_cosmo.py`. The $R + R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); only the proton/electron ratio and the ambient projective limit (G6) are left open by design — no script "proves" those, consistent with their **[A]** marking. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).